SHAPE ROTATOR HACKATHON · IC3 RESEARCH

Elastic Restaking Networks Applied to *Lending*

The first cross-chain lending protocol with portfolio-aware collateral management. Diversified portfolios get more borrowing power. Concentrated ones get less. Backed by formal proofs.

ElasticLend · March 2026

TRACK: DEFI, SECURITY & MECHANISM DESIGN

THE PROBLEM

$200K in collateral. Same value. Very different risk.

ALICE: CONCENTRATED

$200K

100% in ETH across two chains. Looks diversified. ETH crashes 50%, she loses $100K. Underwater.

BOB: DIVERSIFIED

$200K

Split across ETH, BTC, USDC, LINK on four chains. ETH crashes 50%, he loses $25K. Still healthy.

Every lending protocol gives them similar borrowing power. This is wrong.

THE RESEARCH

Elastic Restaking Networks

Roi Bar-Zur and Ittay Eyal, ACM CCS 2025

> *Elastic networks exhibit superior robustness compared to previous approaches, and demonstrate a synergistic effect where an elastic restaking network enhances its blockchain's security.*
>
> COROLLARY 1: ELASTIC NETWORKS ARE STRICTLY MORE EXPRESSIVE THAN ATOMIC ONES.

THE INSIGHT

Lending portfolios are restaking networks

ELASTIC RESTAKING	ELASTICLEND
Validator v	Borrower u
Service s	Risk factor (ETH, BTC, STABLE, OTHER)
Stake σ(v)	Total collateral value
Allocation w(v,s)	Exposure to risk factor
Restaking degree	Concentration degree (HHI)
Attack on service	Market crash
Elastic slashing	Stretch effect after liquidation

Same math. Same theorems. New application.

THE MECHANISM

sqrt of sum of squares

CORE FORMULA

$$\text{Rigid: } BP = \text{Collateral} - \Sigma(\text{loss}_g)$$

$$\text{Elastic: } BP = \text{Collateral} - \sqrt{(\Sigma\ \text{loss}_g^2)}$$

RIGID (AAVE)

$100K

Alice: all ETH

ELASTIC (OURS)

$155K

Bob: diversified

+54% borrowing power for diversified portfolios

CRASH SIMULATION

ETH crashes 40%. Who survives?

RIGID MODEL	ELASTIC MODEL
0.78	1.00
LIQUIDATED	SURVIVES

Scenario: Bob's diversified portfolio, 40% ETH crash, same borrowed amount

Same portfolio. Same crash. Rigid is liquidated. Elastic absorbs the shock.

THE STRETCH EFFECT

After partial liquidation, collateral stretches

BEFORE LIQUIDATION		AFTER LIQUIDATION
72%	→	83%
BP per dollar of collateral		BP per dollar of collateral

"The most impaired collateral is seized first. The remaining portfolio rebalances toward safer assets. The elastic model recognizes this improved risk profile. The collateral stretched."

CROSS-CHAIN ARCHITECTURE

Deployed on one chain. Knows about all chains.

Ethereum
Escrow: aETH, wstETH

Arbitrum
Escrow: GMX, Vaults

Optimism
Escrow: USDC Vaults

Base (Hub)
Elastic Protocol

Tiered liquidation: local first (trustless), then cross-chain (oracle-mediated backstop). Cross-chain diversification is where elastic matters most.

TEST RESULTS

27 tests. All passing.

+54% DIVERSIFICATION BONUS	**6** SMART CONTRACTS	**27** TESTS PASSING
1.00 ELASTIC HF (POST-CRASH)	**0.78** RIGID HF (POST-CRASH)	**2** CHAINS (HUB + ESCROW)

Full end-to-end: escrow deposit, oracle attestation, borrow, crash, tiered liquidation, stretch effect.

THE VISION

What's next for ElasticLend

01 Deploy elastic collateral model to mainnet as a standalone protocol

02 ElasticLend SDK: any lending protocol can integrate portfolio-aware risk

03 Bridge-agnostic cross-chain verification (CCIP, LayerZero, Hyperbridge)

04 Backtest engine with real Aave positions against historical crashes

05 $200B in DeFi lending uses rigid collateral factors. We fix that.

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